EXAMINATION

Luleå University of Technology

Course: **D7024E**
Course name: **Distributed and Mobile Computing Systems**
Date: **2019-10-30**
Time: **10.00-14.00 , online home exam**
Aid: **WWW (no interactive communication with others)**

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Grade scale: **50% grade 3, 75% grade 4 and 90% grade 5**	Note that this grade is for the exam. The exam is supposed to be easy enough so that anyone that followed the course properly should pass it. The grade for the course (including all assessment criteria) may be different.
Total number of questions and score: **3 questions, total score 32**	
Other information: This is an individual home exam.	Note: No interactive cooperation is allowed. Information can be searched on the world wide web. Please provide references, when such information is used in you answers.

General instructions

Check that you have received all the tasks/questions. You shall formulate answers with your own words, but for parts of the answers you may cite sources on the web. In any case you must provide the references to your primary sources (e.g., URL).

After examination

Examination results are posted in Canvas within 20 workdays after the examination.

1. **Mixed question on distributed systems (11p)**

   These are multi-choice questions, with three alternatives. Please answer "true", "false" or “depends” to each of the following statements. If your answer is “depends” you must provide a short explanation to get credits. For “true” or “false”, no explanation is required, unless you want to explain what conditions are assumed. If you have identified a corner case and provide a good explanation, we can accept an answer that is different from the expected one. Each question is worth 0.5p.

   a. A distributed system is always scalable.

   b. In a distributed system, each node (e.g., CPU) has its own local memory and clock.

   c. Marshaling (a k a serialization) is an essential part in providing inter-platform remote procedure calls (RPC).

   d. Kademlia routing is done in linear time O(n), where n is the number of nodes.

   e. The number of k buckets in Kademlia routing depends on the number active nodes in the system.

   f. Distributed Hash Tables (DHTs) often stores immutable data objects, but not immutable history (i.e., ordering) of the objects.

   g. In traditional SQL databases the transaction log (or command log) is typically used locally to handle transaction processing and to support a RAM cache.

   h. In SQL databases there is a need for a transaction log (or command log) to handle and resolve deadlocks.

   i. Many blockchain systems are based on a globally synchronized and replicated log of the entire command history since the inception/instantiation of the system.

   j. Blockchain solution typically scale out linearly, i.e., increase the provided performance linearly with the number of added nodes.

   k. Leader based global consensus solutions typically assume that there can be a small number of traitors.

   l. Leader based global consensus is an efficient way of providing non-Byzantine consensus.

   m. NAT/firewall traversal by TURN may enable communication through a symmetric firewall.

   n. Modern DNS hosting services can support failover, by heart-beating to the

servers providing a service and advertising only live server(s).

o. The pubsub paradigm is based on mutable messages.

p. A pubsub service is in itself typically implemented as a scalable distributed system that shards channels or topics across servers.

q. Kafka does not need any internal global state synchronization.

r. Safety and liveness must be ensured in order to reach global consensus.

s. Zookeeper is a good solution for collaborating nodes to record and keep track of whether each of them is alive/reachable or not.

t. Map-reduce is designed to move computation to where data is stored.

u. PGP uses public/private key pairs to encrypt the payload data sent between parties.

v. Certificate authorities are trusted entities that certify the ownership of public keys.

Note: In the following questions 2 and 3, it is important to include sources (e.g., URLs) and that you explain the answer in you own words.

2. Microservices (12p)

a) What is the purpose of service discovery? Explain how service discovery is typically provided in the microservices approach. (4p)
b) What is the purpose of an event handler/bus in microservices? (4p)
c) How is database synchronization typically handled, and why is database consistency a challenge in microservices? (4p)

3. Blockchain (13p)

a) In Bitcoin, explain schematically the role, use, value and validity of a coinbase transaction (5p)
b) In Ethereum, how are multiple smart contracts executed and verified by a large number of nodes, in order to obtain consensus on the correctness of each new block and consensus on the final state? (8p)